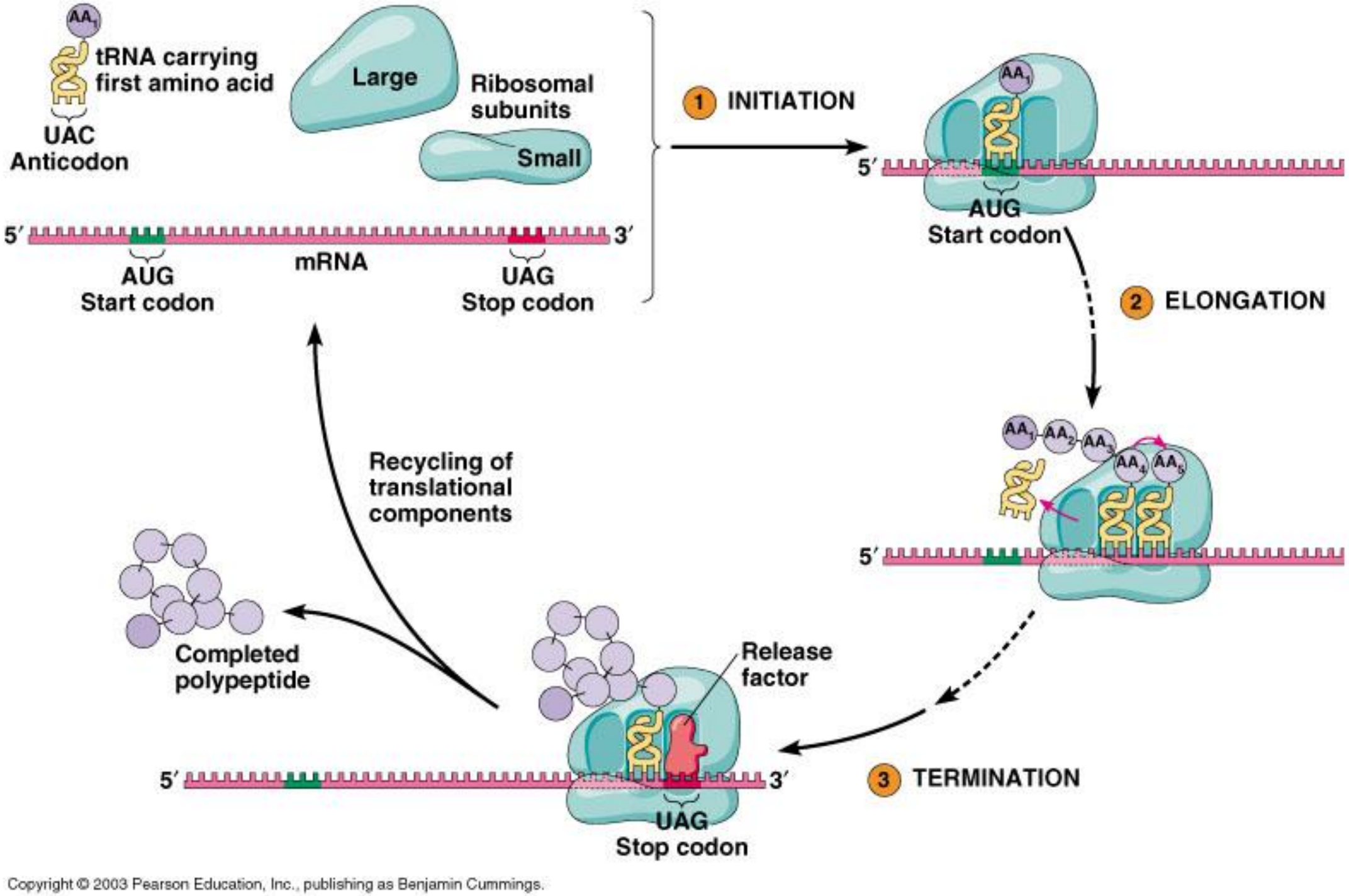
Attendance QR code

Translation: basic mechanism

amino acyl tRNA synthetase

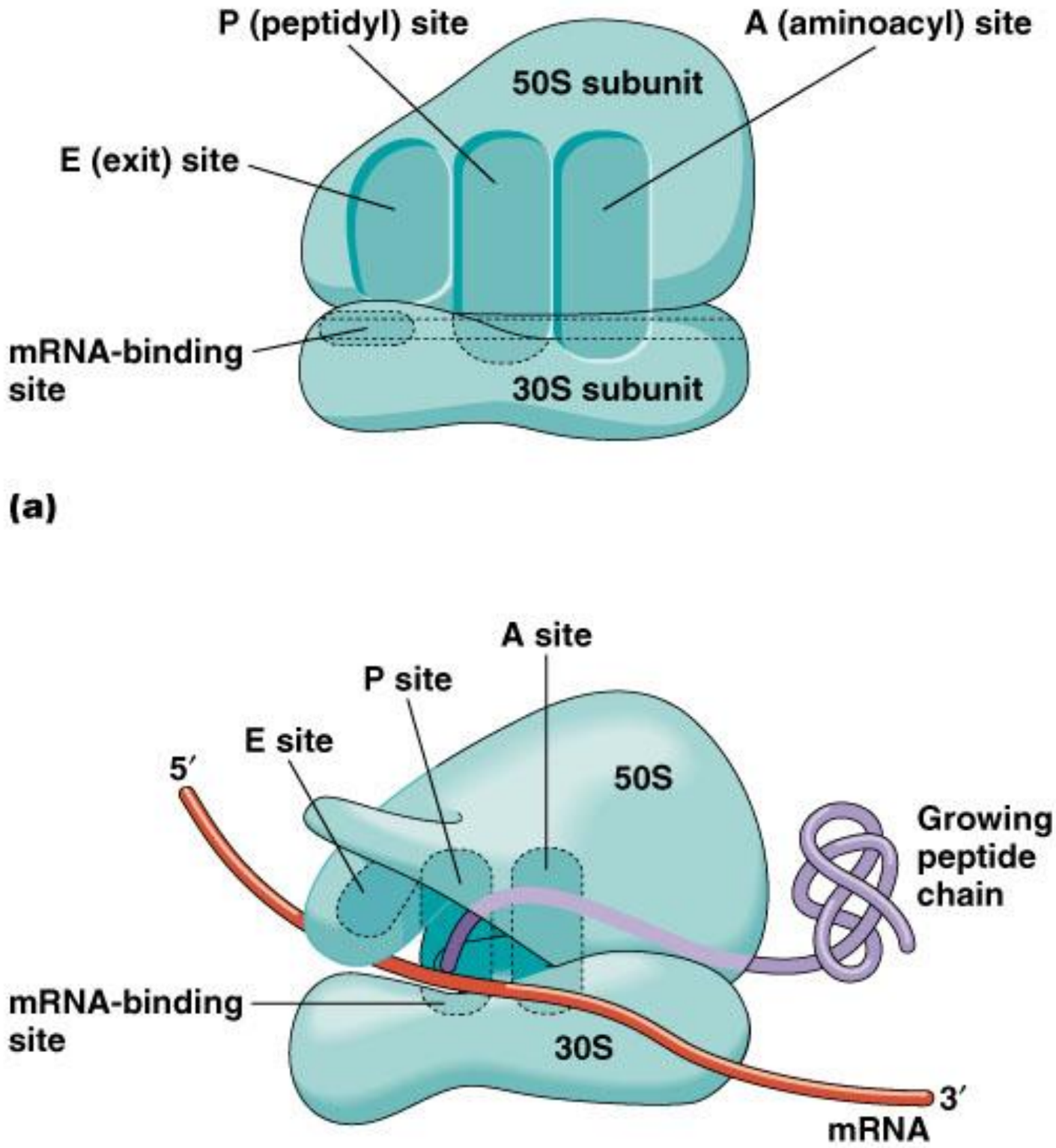


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Ribosomes synthesise proteins at a rate of 3 - 5 amino acids added per second (It takes less than a minute to make a small, 100-200 aa protein)

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Ribosome sites



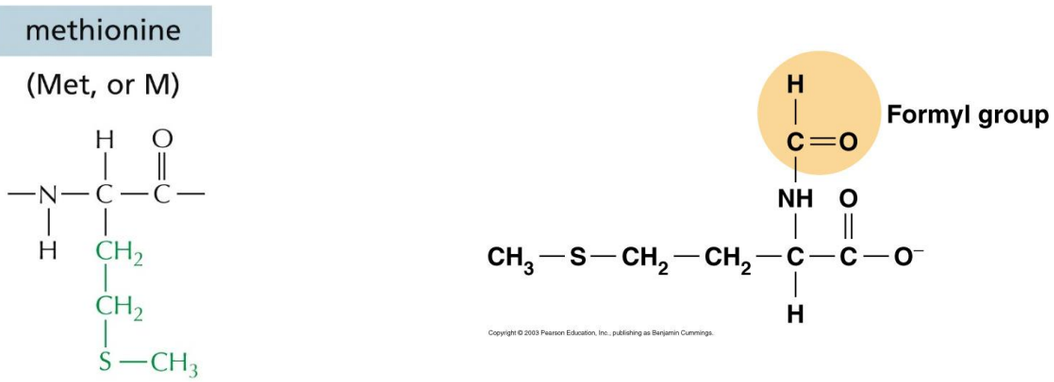
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Where does translation start?

- Translation starts at the initiator methionine (AUG)

- Synthesis of all protein chains in prokaryotic and eukaryotic cells begins with the amino acid Methionine (Met, or M) (f-Met in prokaryotes)



Methionine residues are also present within the protein

Q: How does the ribosome know which methionine is the initiator methionine?

A: There are two tRNAs for methionine-one is the Met initiator tRNA and the other is the 'regular' Met tRNA. The initiator tRNA has a different nucleotide sequence from regular tRNA. It cannot interact with elongation factors and vice versa.

Translation initiation in prokaryotes

- In prokaryotes, the 16S rRNA recognises and base-pairs with a nucleotide sequence in the mRNA (Shine-Dalgarno sequence) near the initiator AUG

- bacterial rRNA recruits the ribosomal subunit to the translation start site

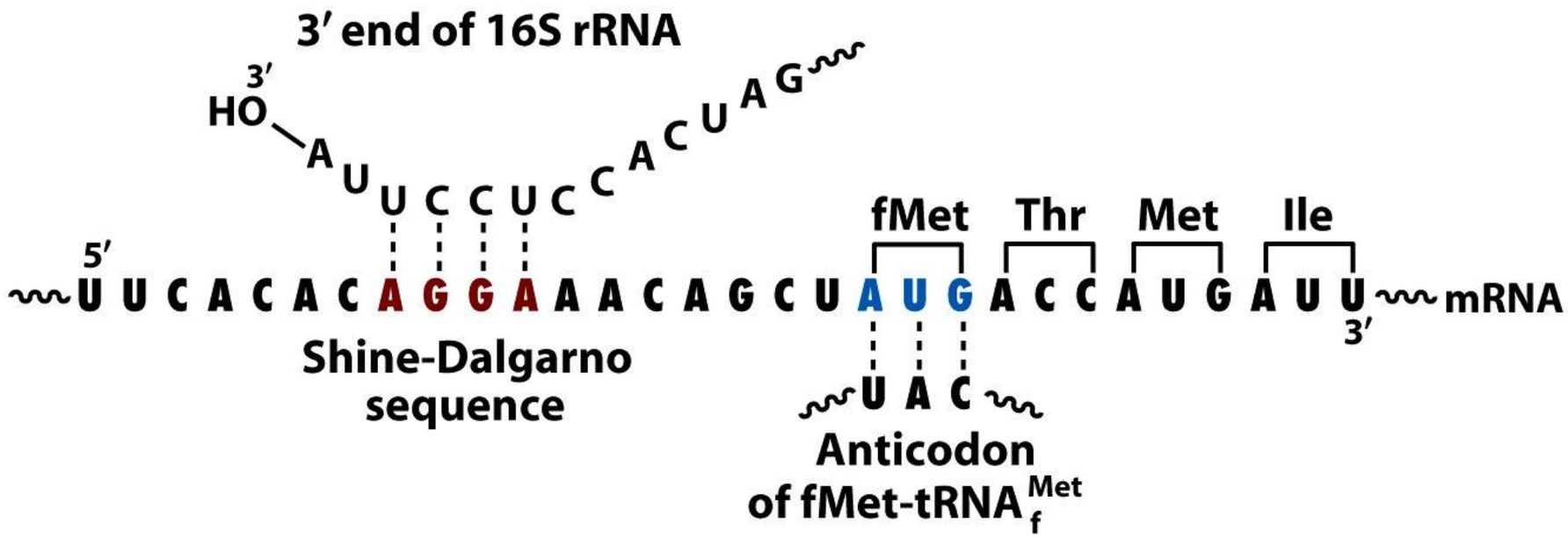


Figure 22-17b Principles of Biochemistry, 4/e
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Translation initiation in eukaryotes-locating where to start

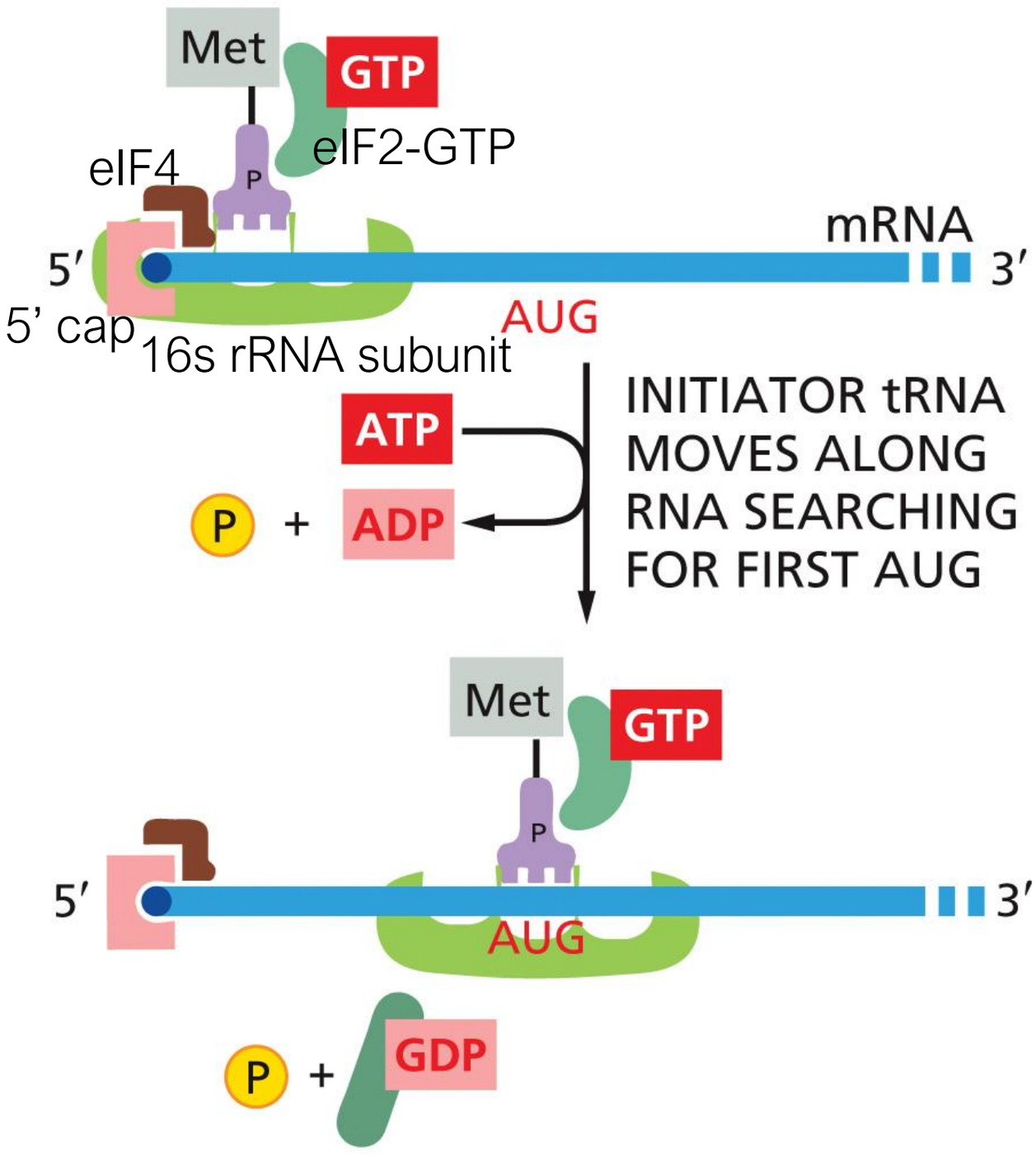


Figure 6–74 (Part 2) Molecular biology of the cell 7e:The initiation of protein synthesis in eukaryotes

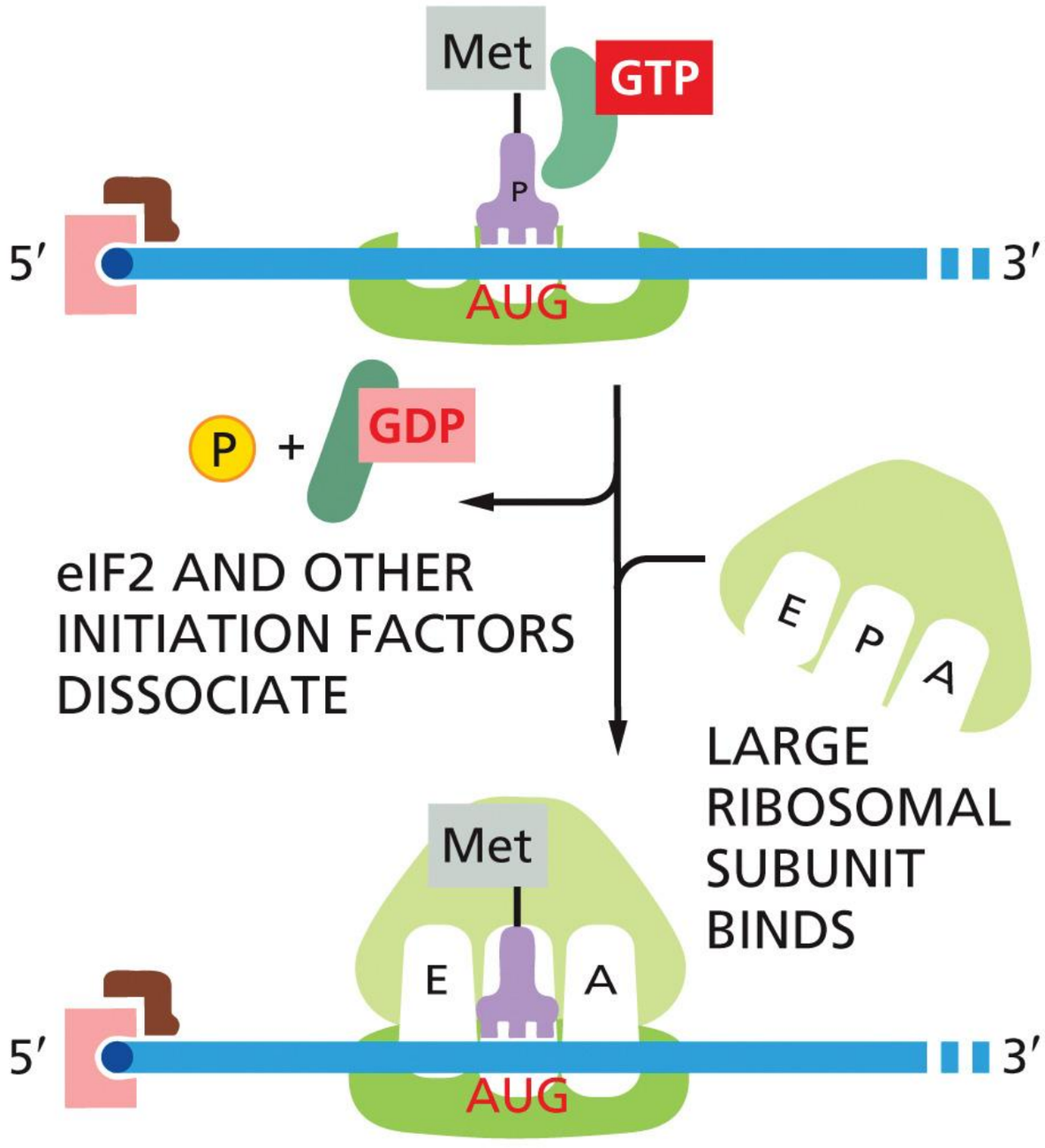
.

- Marilyn Kozak - analysed 699 eukaryotic mRNA sequences to identify the translation initiation consensus

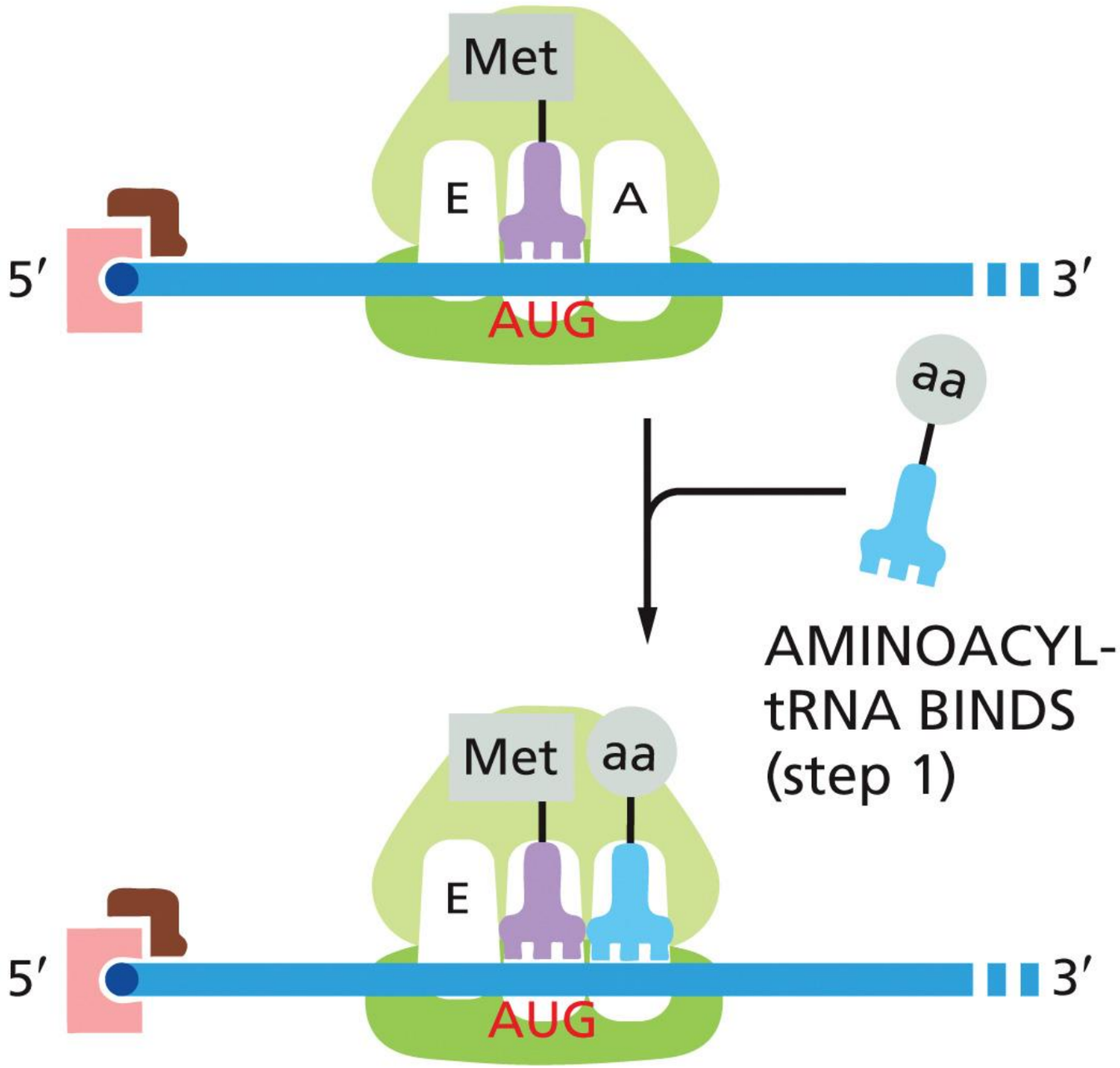
- In eukaryotes, the ribosome binds to the mRNA near the 5’ capped end and translation starts at the <u>Kozak sequence:</u>

mRNA 5’-GCC(G or A)CC**<u>AUG</u>**G-3’

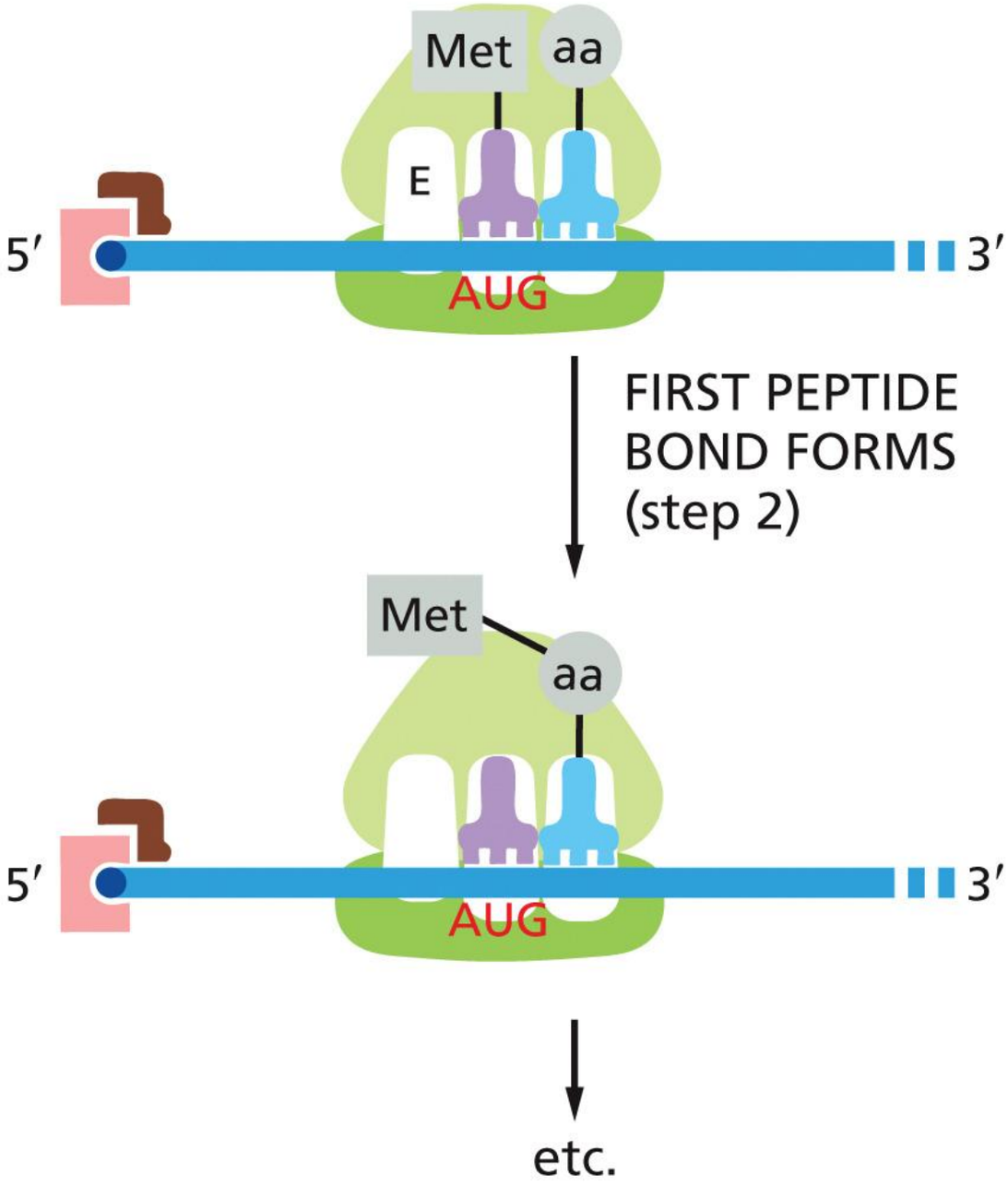
Translation initiation in eukaryotes-completing the ribosome



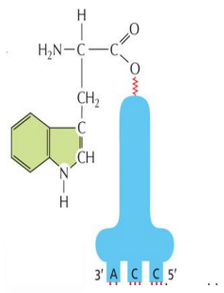
Translation initiation in eukaryotes-the second tRNA binds



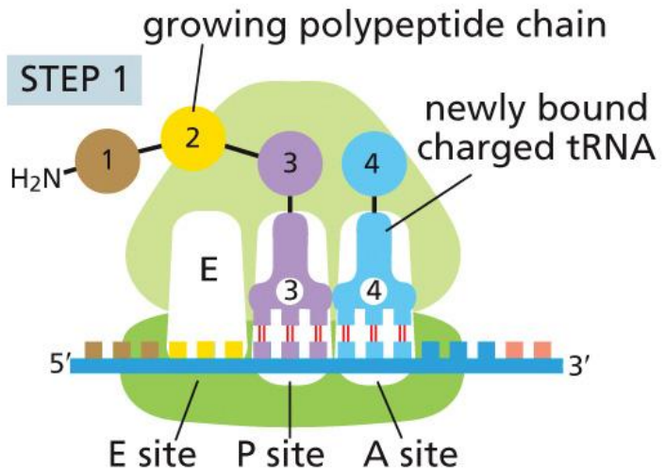
Translation initiation in eukaryotes-first peptide bond formation

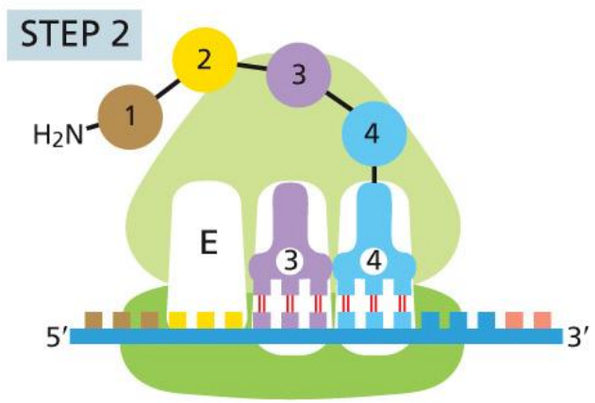


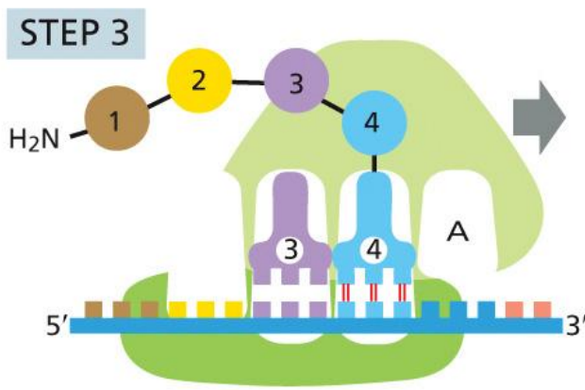
Initiation phase is followed by elongation phase

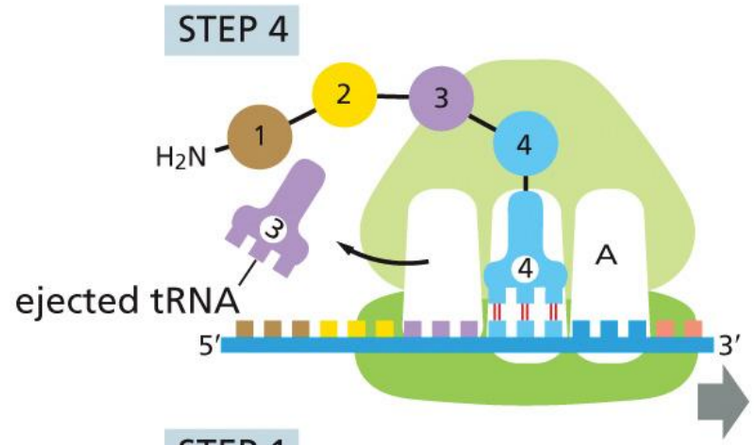


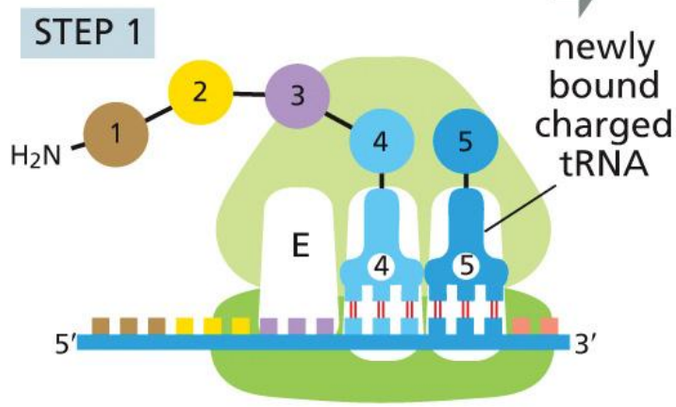
Amino acyl tRNA



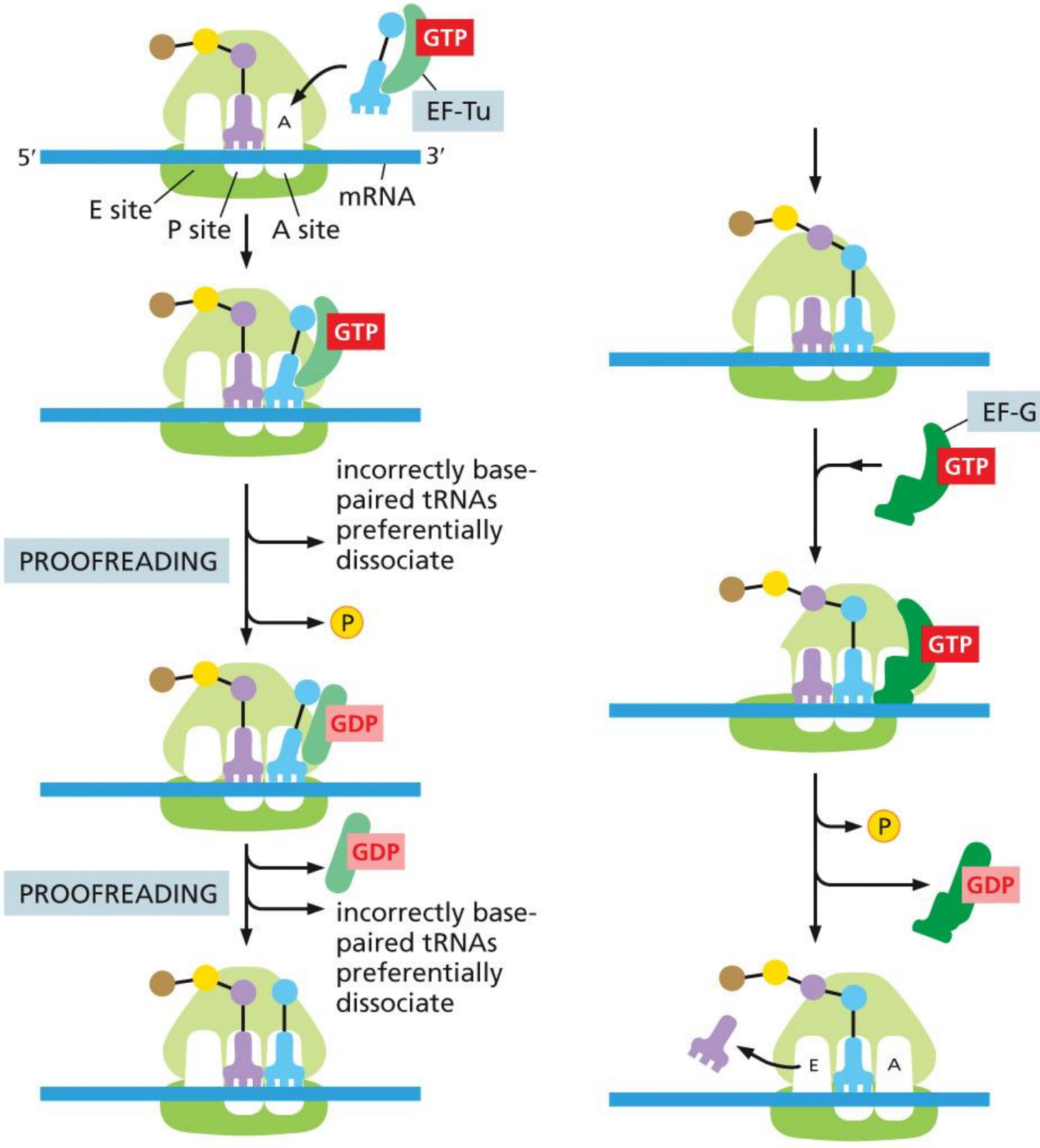






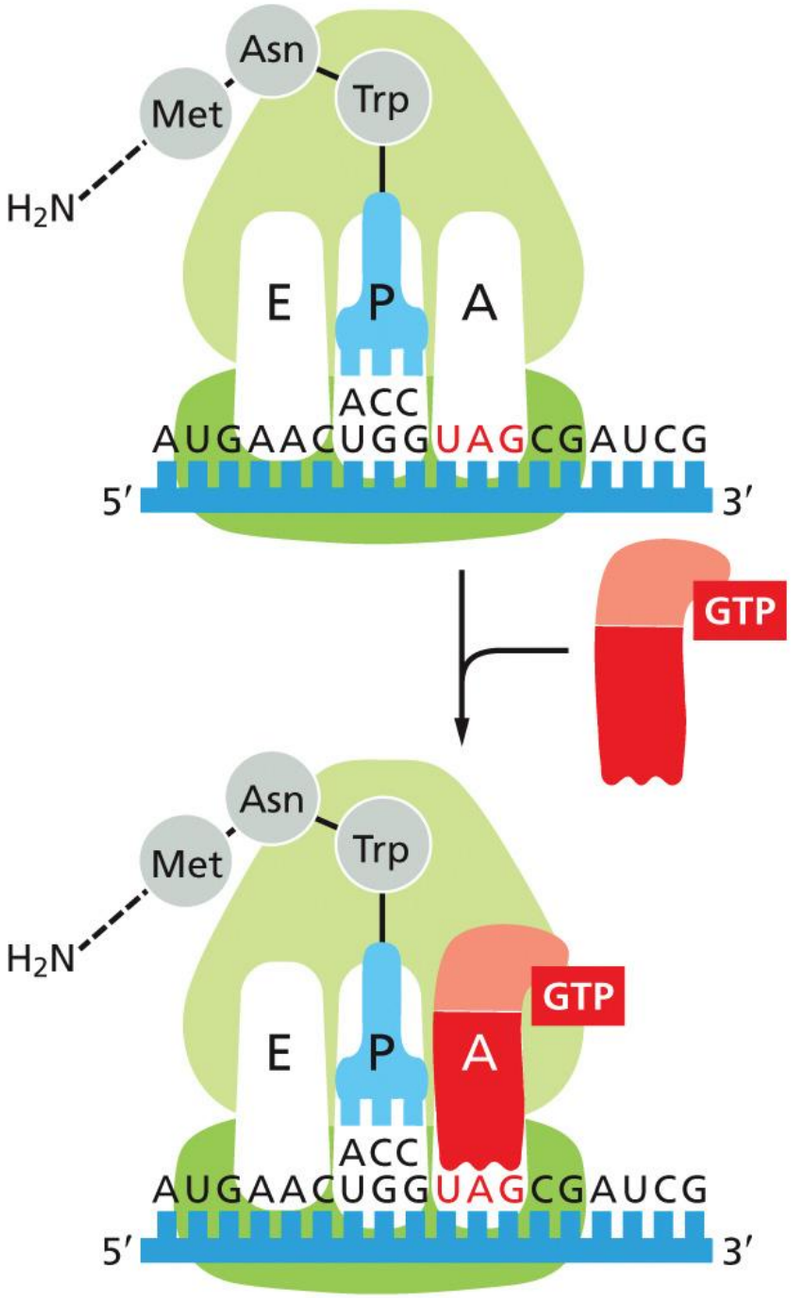


Elongation factors and proofreading



EF-Tu and EF-G in bacteria,
EF1 and EF2 in eukaryotes

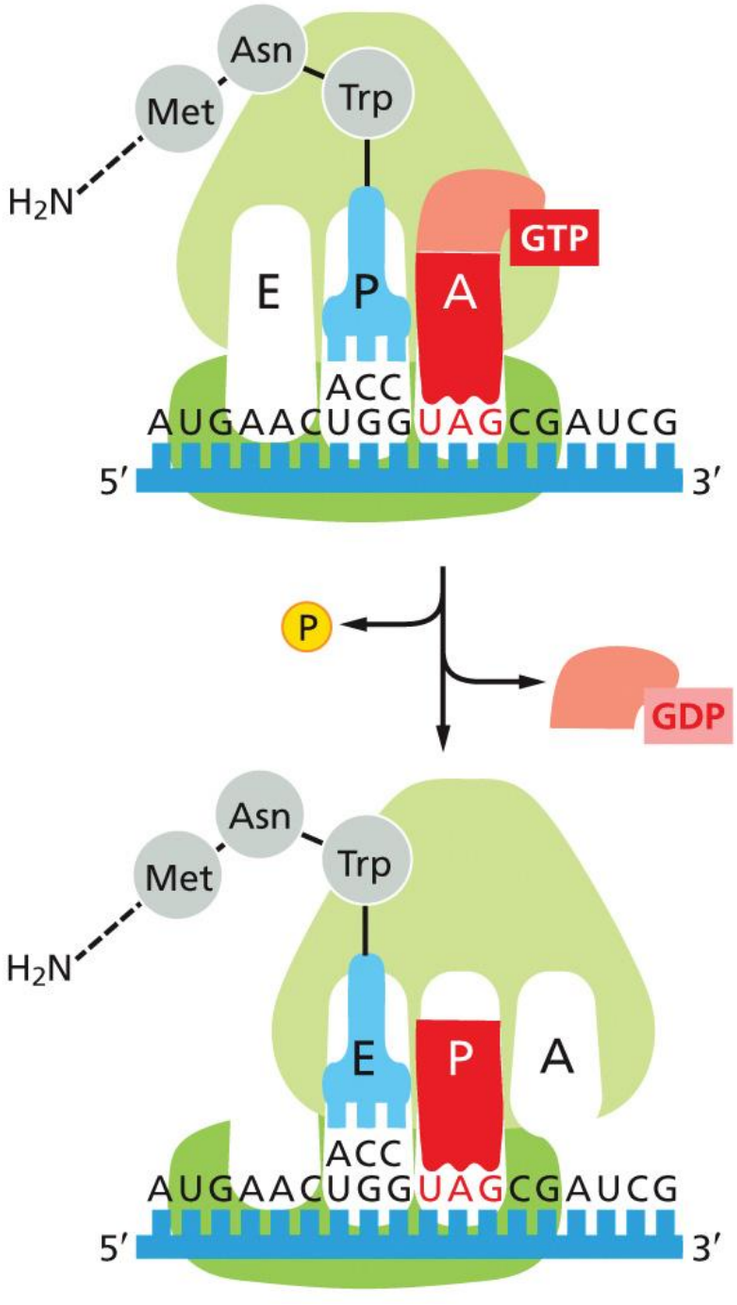
Termination phase



In almost all cells, UAA, UGA and UAG specify stop (translation termination) signals

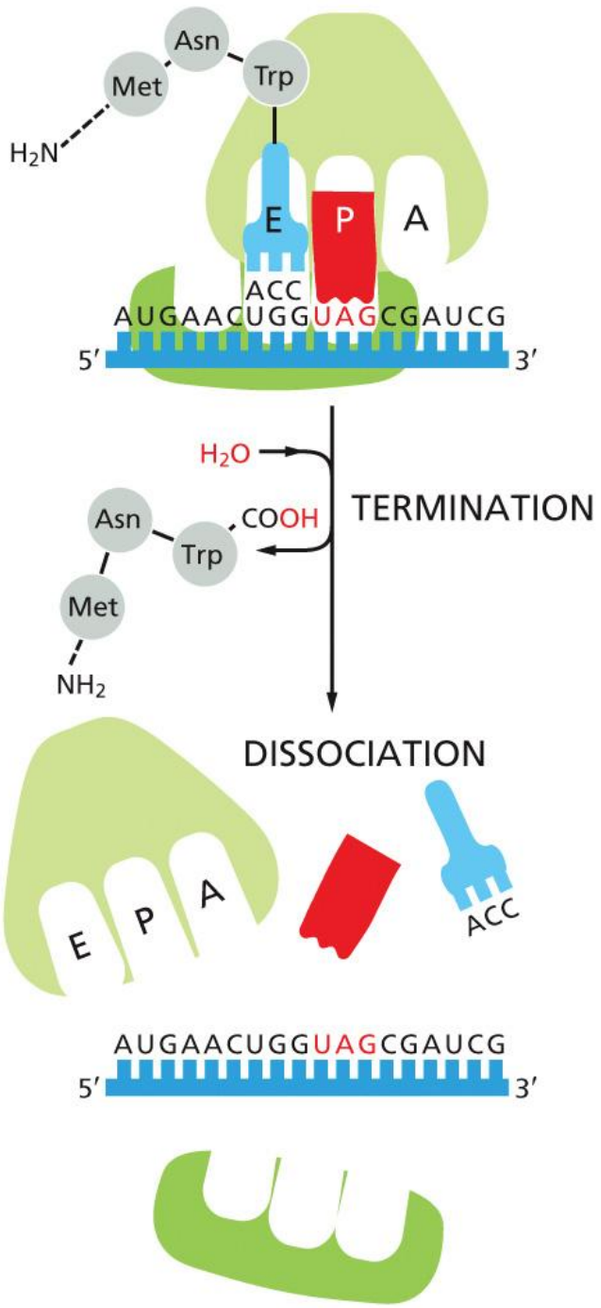
Release factors are proteins shaped like tRNA-molecular mimicry, that recognise the stop codon and bind to it.

Termination phase



Hydrolysis of GTP facilitates translocation so that the release factor is in the P site

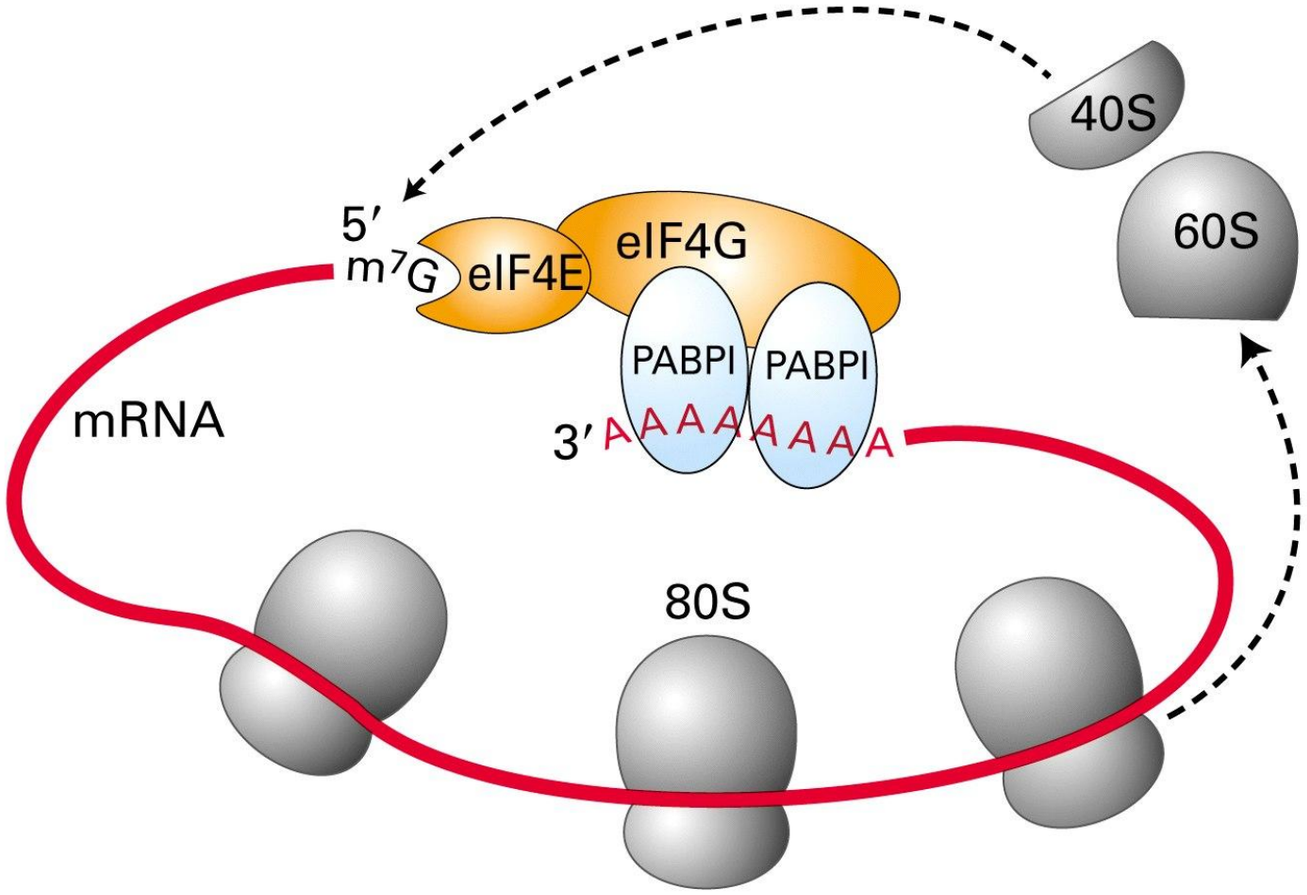
Termination phase

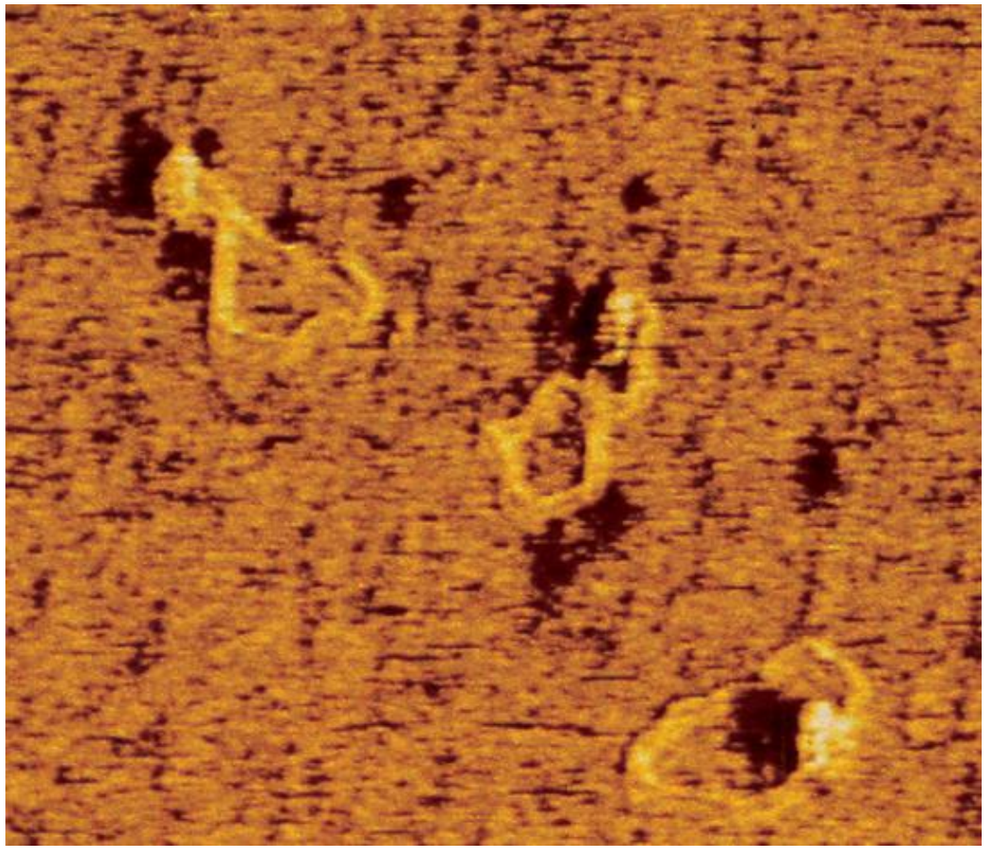


- Prokaryotes:
- RF1 recognises UAG/UAA
- RF2 recognises UGA/UAA
- RF3- hydrolyses the peptidyl tRNA in the P site which results in release of the peptide chain and RF1/2 from the ribosome.

- Eukaryotes:
- eRF1-recognises all stop codons
- eRF-3-promotes cleavage of peptidyl-tRNA by hydrolysis.

The polyA tail

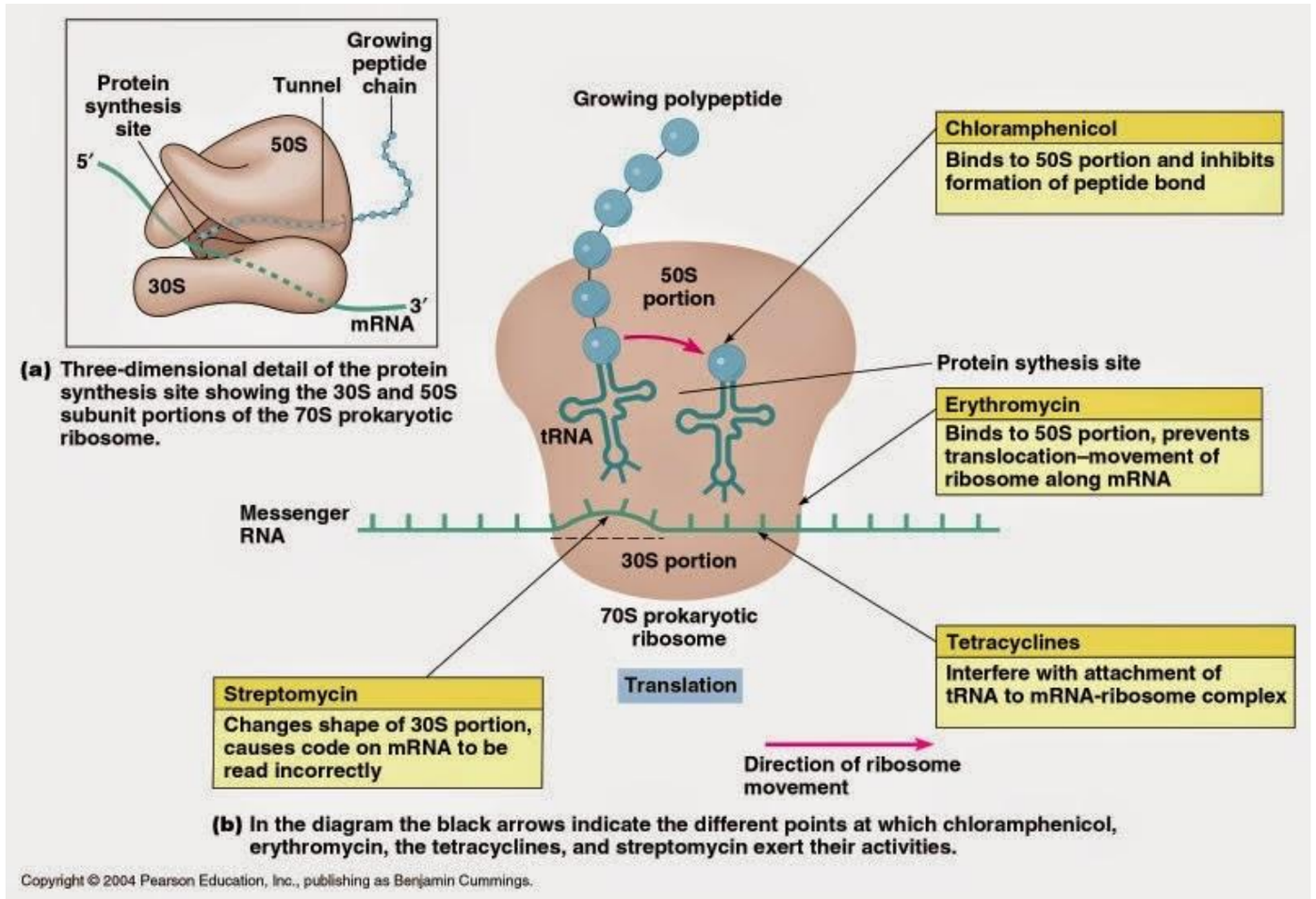




Force-field EM showing circular mRNA

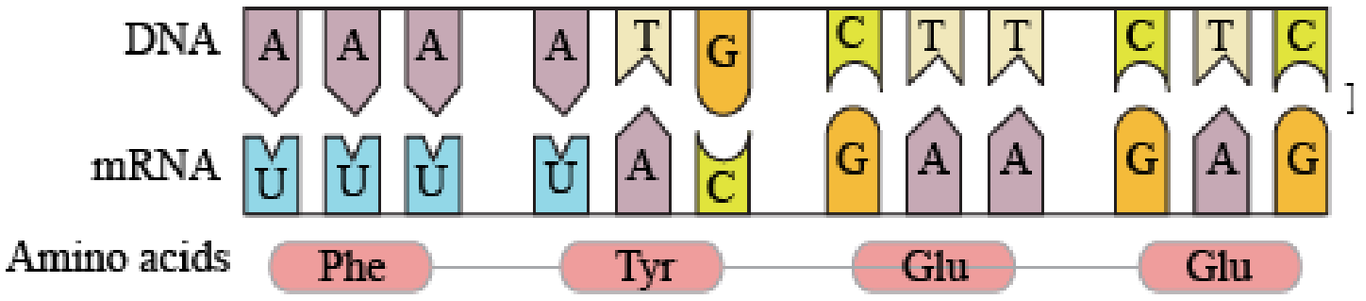
Inhibitor	Specific effect
Acting only on bacteria	
Tetracycline	Blocks binding of aminoacyl-tRNA to the A site of ribosome
Streptomycin	Prevents the transition from translation initiation to chain elongation and also causes miscoding
Chloramphenicol	Blocks the peptidyl transferase reaction on ribosomes (step 2 in Figure 6–64)
Erythromycin	Binds in the exit channel of the ribosome and thereby inhibits elongation of the peptide chain

Table 6.4 Molecular biology of the cell 6e



https://1.bp.blogspot.com/-KC9tsVM9LJU/U7t-coW5LsI/AAAAAAAAAJY/dTfMi273k30/s1600/2.jg

The genetic code: codons and reading frame



A **triplet** code - 64 codons.(4x4x4=64)

Reading Frame



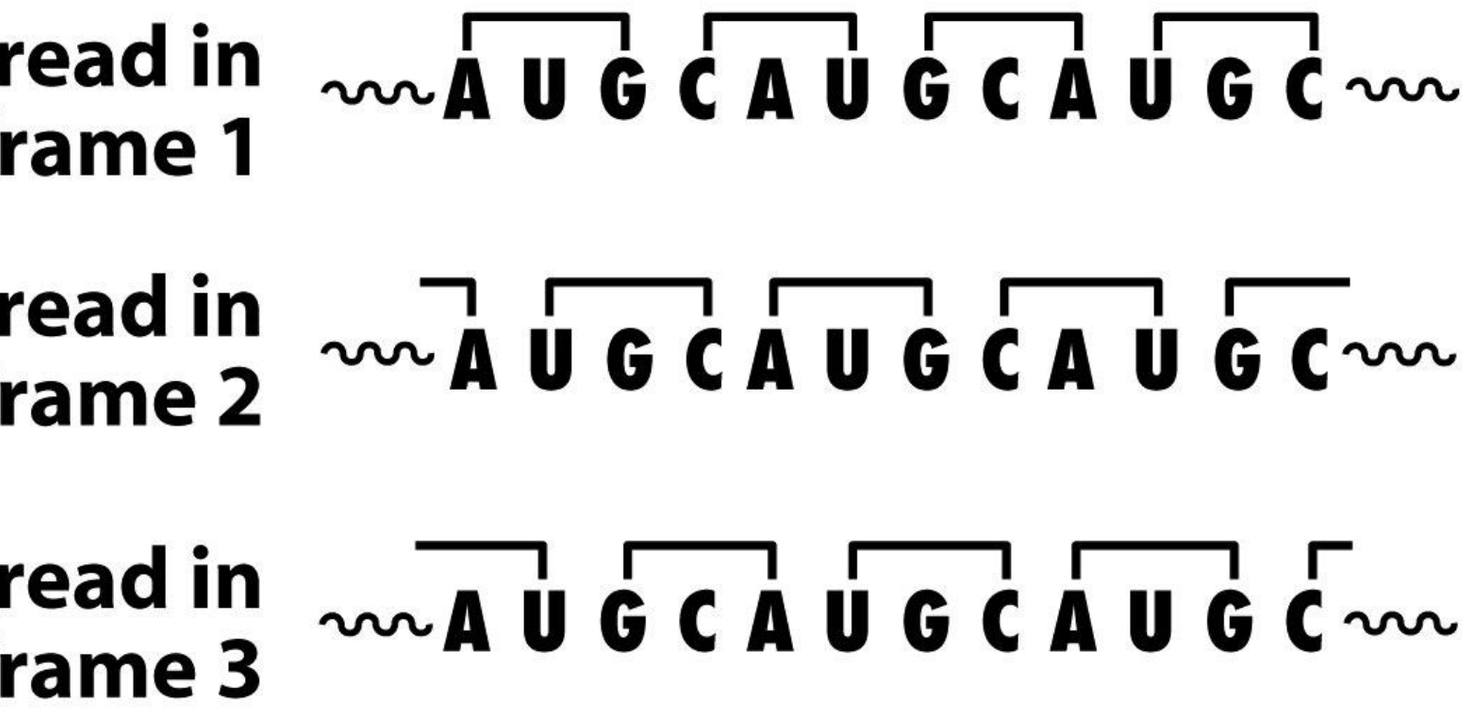


Figure 22-2 Principles of Biochemistry, 4/e
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- The code is **degenerate**: more than one codon can be used to specify an amino acid
- 61 codons specify amino acids, but there are only 20 amino acids used in proteins and encoded directly in DNA.

First position (5′ end)	*Second position* U	C	A	G	*Third position (3′ end)*
U	Phe	Ser	Tyr	Cys	U
	Phe	Ser	Tyr	Cys	C
	Leu	Ser	STOP	STOP	A
	Leu	Ser	STOP	Trp	G
C	Leu	Pro	His	Arg	U
	Leu	Pro	His	Arg	C
	Leu	Pro	Gln	Arg	A
	Leu	Pro	Gln	Arg	G
A	Ile	Thr	Asn	Ser	U
	Ile	Thr	Asn	Ser	C
	Ile	Thr	Lys	Arg	A
	Met	Thr	Lys	Arg	G
G	Val	Ala	Asp	Gly	U
	Val	Ala	Asp	Gly	C
	Val	Ala	Glu	Gly	A
	Val	Ala	Glu	Gly	G

Figure 22-3 Principles of Biochemistry, 4/e
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- mRNAs can be correctly translated by the protein synthesizing machinery of very different species.
- E.g. human haemoglobin mRNA is correctly translated by wheatgerm extract
- Bacteria (E.Coli) efficiently express recombinant DNA molecules encoding human protein such as insulin.

There are 61 codons

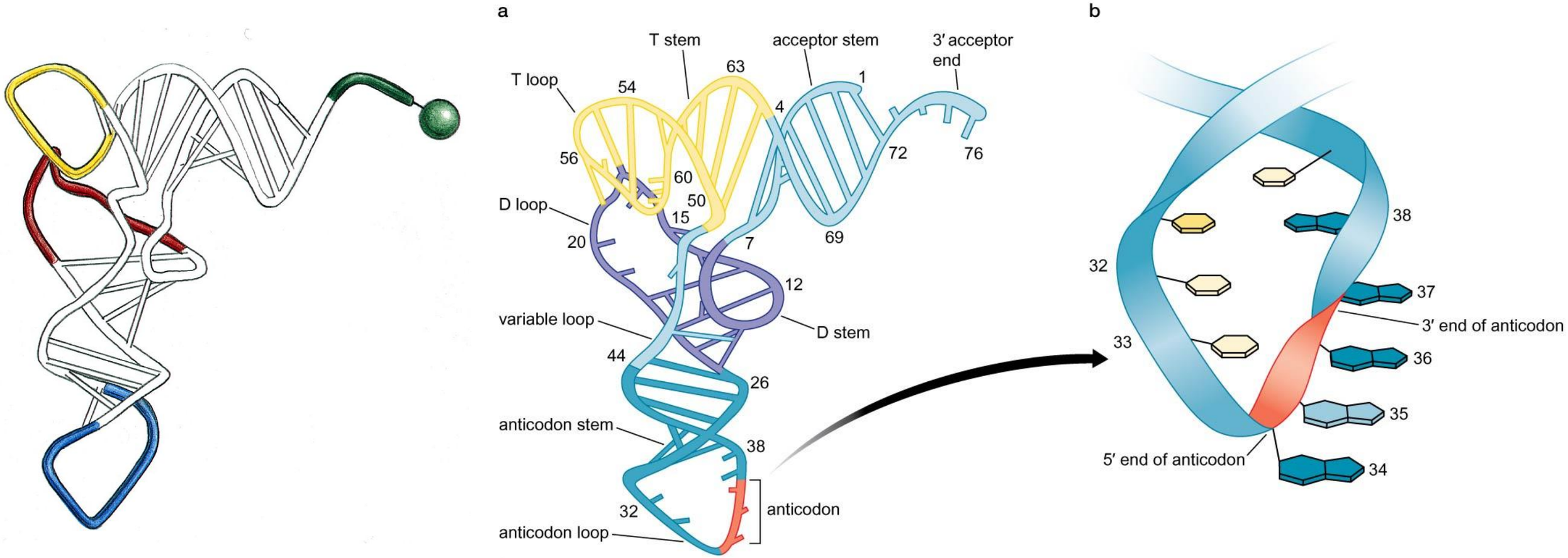
Are there 61 tRNA molecules with matching anticodons?

Not in most cells-humans have tRNAs with only 48 anticodons, and the number varies greatly between species

How do fewer than 61 tRNAs recognise the 61 codons?

- There is more than one codon for each amino acid
- There is more than one tRNA for some amino acids

A single tRNA can recognise more than one codon corresponding to a given amino acid



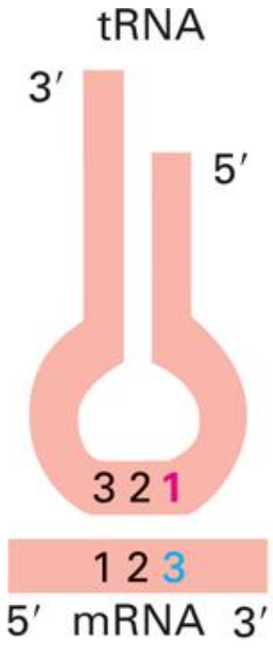
Green: amino accepting arm
Blue: anticodon arm

Figure 16.3 Molecular biology of the gene 7e Watson.

A single tRNA can recognise more than one codon corresponding to a given amino acid

This is as a result of the **wobble** phenomenon, where non-standard base pairing can occur

-it usually occurs in the third base position of the mRNA (*the first base of the anticodon in the tRNA*)



If these bases are in first, or wobble, position of anticodon					
C	A	G	U	I	
G	U	C U	A G	C A U	then the tRNA may recognize codons in mRNA having these bases in third position



If these bases are in third, or wobble, position of codon of an mRNA				
C	A	G	U	
G I	U I	C U	A G I	then the codon may be recognized by a tRNA having these bases in first position of anticodon

Figure 4.21
MCB Lodish 7e

Translation: take home message

- Ribosomes are large ribonucleoproteins comprised of two subunits containing several different rRNAs and more than 50 proteins

- The catalytic activity is provided by the large RNA s ubunit(23s/28s)

- As with transcription, there is an initiation, elongation and termination phase. Each of these is highly regulated by initiation/elongation/termination factors (regulatory proteins).

- They read the genetic code from mRNA in triplet codons.

- There is degeneracy in the genetic code meaning that multiple codons can encode the same amino acid, and multiple tRNAs can read one codon.

-